# PROJECT REPORT

# ANALYZE THE HEALTHCARE COST AND UTILIZATION IN WISCONSIN HOSPITALS

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# BUSINESS SCENARIO

A nationwide survey of hospital costs conducted by the US Agency for Healthcare consists of hospital records of inpatient samples. The given data is restricted to the city of Wisconsin and relates to patients in the age group 0-17 years. The agency wants to analyse the data to research on the healthcare costs and their utilization.

Attributes Description:

AGE - Age of the patient discharged

FEMALE - Binary variable that indicates if the patient is female

LOS - Length of stay, in days

RACE - Race of the patient (specified numerically)

TOTCHG - Hospital discharge costs

APRDRG - All Patient Refined Diagnosis Related Groups

# TASK TO BE PERFORMED

1) To record the patient statistics, the agency wants to find the age category of people who frequent the hospital and has the maximum expenditure.

2) In order of severity of the diagnosis and treatments and to find out the expensive treatments, the agency wants to find the diagnosis related group that has maximum hospitalization and expenditure.

3) To make sure that there is no malpractice, the agency needs to analyse if the race of the patient is related to the hospitalization costs.

4) To properly utilize the costs, the agency has to analyse the severity of the hospital costs by age and gender for proper allocation of resources.

5) Since the length of stay is the crucial factor for inpatients, the agency wants to find if the length of stay can be predicted from age, gender, and race.

6) To perform a complete analysis, the agency wants to find the variable that mainly affects the hospital costs.

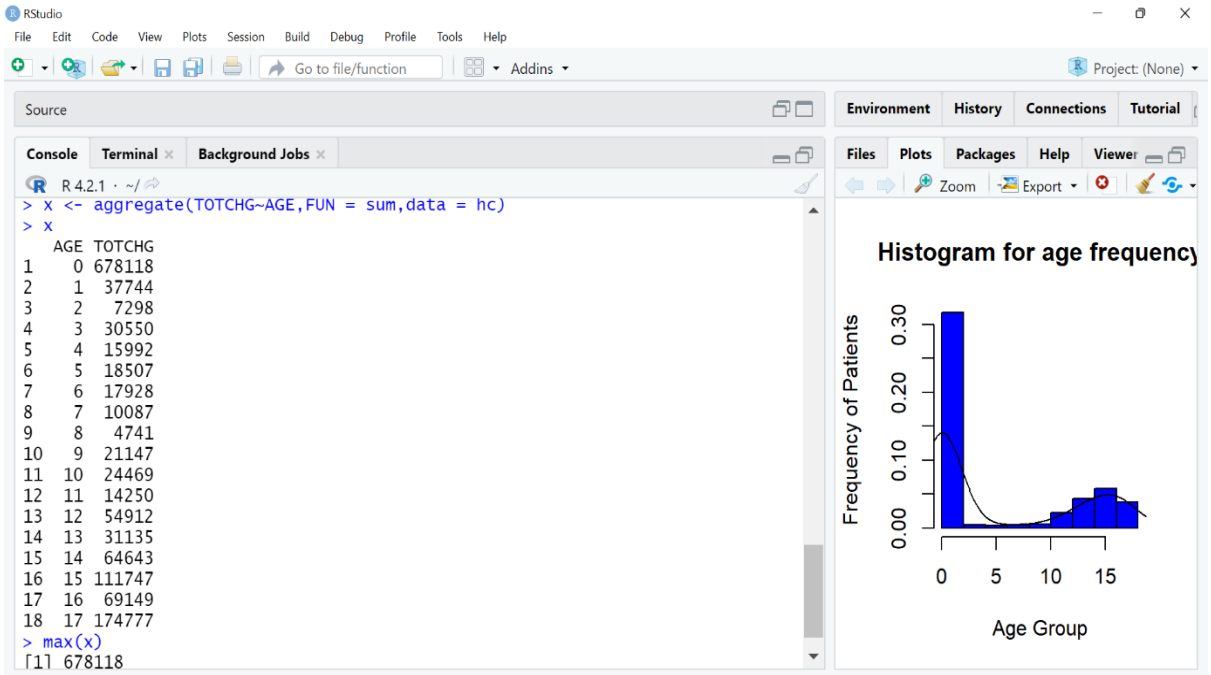
# CODE

```
#Project on Healthcare
#importing file
library(readxl)
hc <- read.csv(file.choose())
View(hc)
summary(hc)

#part1
hist(hc$AGE, main = "Histogram for age frequency",
   xlab = "Age Group",
   ylab = "Frequency of Patients",
   prob= TRUE,
   col = "blue")
lines(density(hc$AGE))
summary(as.factor(hc$AGE))
x <- aggregate(TOTCHG~AGE,FUN = sum,data = hc)
x
max(x)

#part2
which.max(summary(as.factor(hc$APRDRG)))

dc <- aggregate(TOTCHG ~ APRDRG, FUN = sum, data = hc)
dc
dc[which.max(dc$TOTCHG),]


#part3
str(hc)
summary(as.factor(hc$RACE))
```

```
head(hc)
hc<-na.omit(hc)
hc$RACE<-as.factor(hc$RACE)
mod<- aov(TOTCHG ~ RACE, data = hc)
mod
summary(mod)
summary(hc$RACE)

#part 4
mod1 <- lm(TOTCHG ~ AGE + FEMALE, data = hc)
hc$FEMALE<-as.factor(hc$FEMALE)
mod1 <- lm(TOTCHG ~ AGE + FEMALE, data = hc)
summary(mod1)
summary(hc$FEMALE)
head(hc)


#part 5
hc$RACE<-as.factor(hc$RACE)
mod2 <- lm(TOTCHG ~ AGE + FEMALE + RACE, data = hc)
summary(mod2)

#part6
mod3 <- lm(TOTCHG ~ ., data = hc)
summary(mod3)
```

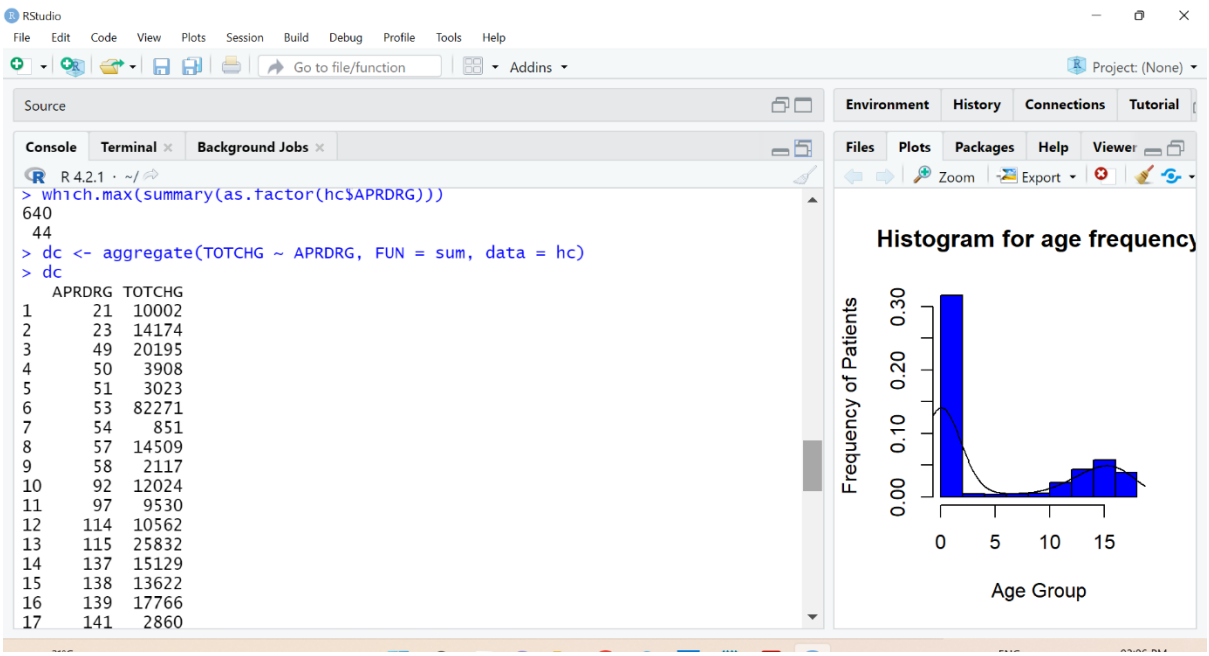
# ANALYSIS

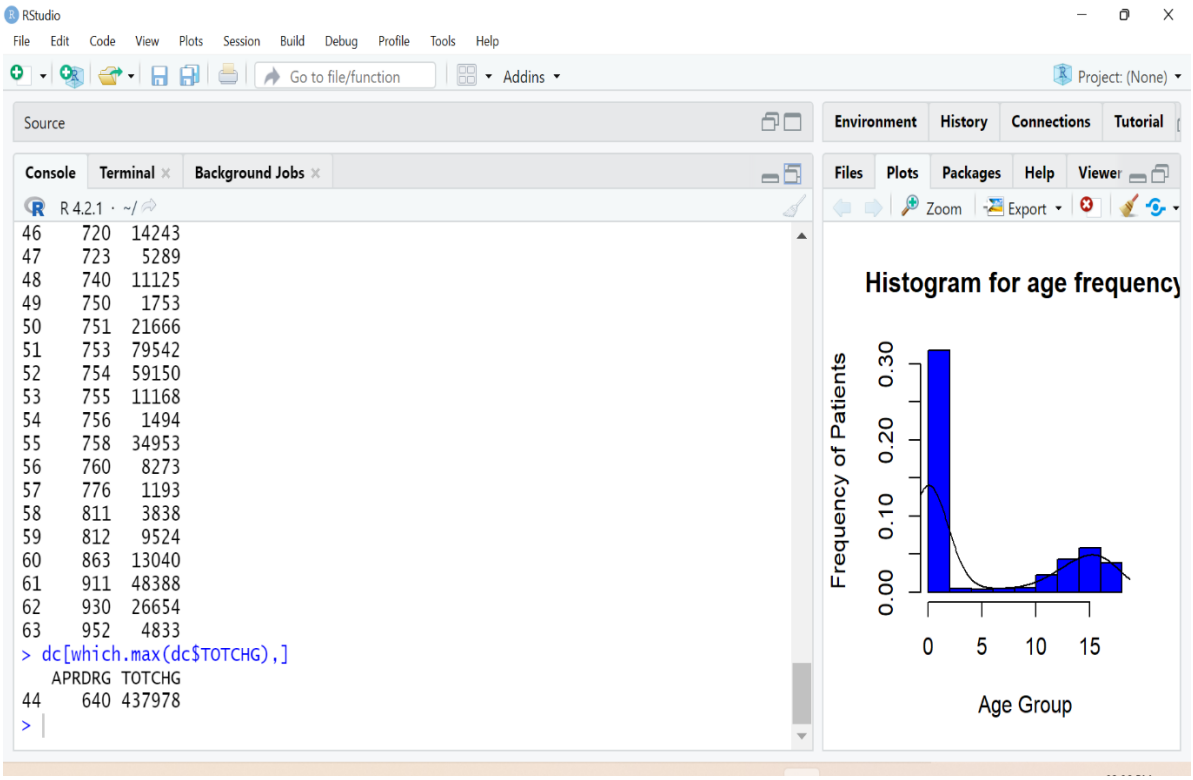
**1) Record the patient statistics, the agency wants to find the age category of people who frequent the hospital and has the maximum expenditure.**




- After executing histogram, we can analyse that age group with most frequency of patients is: 0-1yrs.
- Using as.factor () function, allows us to get the count of patients belonging to different age groups.
- Using aggregate, we can find out the total charges by each age group. To achieve the total amount of all the people of that age group we provide (fun= sum) in aggregate ( ) function.
- Hence,
- Age group with max visits: 0-1 yrs
- Maximum total charges: 678118

2. In order of severity of the diagnosis and treatments and to find out the expensive treatments, the agency wants to find the diagnosis related group that has maximum hospitalization and expenditure.



- Using which.max( ) we get the elements with the biggest value in the vector. Also to get the total maximum expenditure from all the diaganosis groups we use aggregate() function.
- So, diaganosis group with max value : 640
- Max expenditure from group 640 is : 437978

3. To make sure that there is no malpractice, the agency needs to analyze if the race of the patient is related to the hospitalization costs.

```
> summary(as.factor(hc$RACE))
  1   2   3   4   5   6
484   6   1   3   3   2
> head(hc)
  AGE FEMALE LOS RACE TOTCHG APRDRG
1  17      1   2    1   2660    560
2  17      0   2    1   1689    753
3  17      1   7    1  20060    930
4  17      1   1    1    736    758
5  17      1   1    1   1194    754
6  17      0   0    1   3305    347
> hc<-na.omit(hc)
> hc$RACE<-as.factor(hc$RACE)
> mod<- aov(TOTCHG ~ RACE, data = hc)
> mod
Call:
   aov(formula = TOTCHG ~ RACE, data = hc)

Terms:
                    RACE  Residuals
Sum of Squares    18593279 7523518505
Deg. of Freedom          5        493

Residual standard error: 3906.493
Estimated effects may be unbalanced
> summary(mod)
             Df    Sum Sq  Mean Sq F value Pr(>F)
RACE          5 1.859e+07  3718656   0.244  0.943
Residuals   493 7.524e+09 15260687
> summary(hc$RACE)
  1   2   3   4   5   6
484   6   1   3   3   2
```

- The Residual Value (deviation of the observed value) is very high specifying that there is no relation between the race of patient and the hospital cost.
- From the summary we can also see that the data has 484 patients of Race 1 out of the 500 entries.
- This will affect the results of ANOVA as well, since the number of observations is very much skewed.

**4. To properly utilize the costs, the agency has to analyze the severity of the hospital costs by age and gender for proper allocation of resources.**

```
> summary(as.factor(hc$RACE))
  1   2   3   4   5   6
484   6   1   3   3   2
> head(hc)
  AGE FEMALE LOS RACE TOTCHG APRDRG
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> summary(hc$RACE)
  1   2   3   4   5   6
484   6   1   3   3   2
```

- To perform this analysis, we have used Linear Regression model.
- Age is a very important factor in the hospital costs as seen by the significance levels and p-values.
- The gender also seems to have an impact.
- There is an equal number of male and female patients.
- Based the negative coefficient we can conclude that females incur lesser cost than males.

5. Since the length of stay is the crucial factor for inpatients, the agency wants to find if the length of stay can be predicted from age, gender, and race.

```
4  17      1   1    1    736    758
5  17      1   1    1   1194    754
6  17      0   0    1   3305    347
> hc$RACE<-as.factor(hc$RACE)
> mod2 <- lm(TOTCHG ~ AGE + FEMALE + RACE, data = hc)
> summary(mod2)

Call:
lm(formula = TOTCHG ~ AGE + FEMALE + RACE, data = hc)

Residuals:
   Min     1Q Median     3Q    Max 
 -3401  -1449   -874   -135  44955 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  2726.91     265.15  10.284  < 2e-16 ***
AGE            85.43      25.85   3.304  0.00102 ** 
FEMALE1      -746.10     358.25  -2.083  0.03780 *  
RACE2         784.28    1597.85   0.491  0.62376    
RACE3        1060.19    3876.27   0.274  0.78458    
RACE4        -653.67    2240.67  -0.292  0.77061    
RACE5        -700.24    2247.04  -0.312  0.75545    
RACE6       -1688.27    2739.58  -0.616  0.53802    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3865 on 491 degrees of freedom
Multiple R-squared:  0.02761,	Adjusted R-squared:  0.01374 
F-statistic: 1.991 on 7 and 491 DF,  p-value: 0.05456

> mod3 <- lm(TOTCHG ~ ., data = hc)
> summary(mod3)
```

- The significance codes are almost null for all the variables, except for the intercept.
- The p-value high which signifies that there is no linear relationship between the given variables.
- Hence we cannot predict the length of stay of the patients based on the age, gender, and race.

6. To perform a complete analysis, the agency wants to find the variable that mainly affects the hospital costs.

```
RStudio
File  Edit  Code  View  Plots  Session  Build  Debug  Profile  Tools  Help
Source
Console  Terminal  Background Jobs
R 4.2.1 · ~/
Residual standard error: 2622 on 489 degrees of freedom
Multiple R-squared:  0.5544,    Adjusted R-squared:  0.5462
F-statistic:  67.6 on 9 and 489 DF,  p-value: < 2.2e-16

> summary(mod3)

Call:
lm(formula = TOTCHG ~ ., data = hc)

Residuals:
   Min     1Q Median     3Q    Max
 -6367   -691   -186    121  43412

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  5024.9610   440.1366  11.417  < 2e-16 ***
AGE           133.2207    17.6662   7.541 2.29e-13 ***
FEMALE1      -392.5778   249.2981  -1.575    0.116
LOS           742.9637    35.0464  21.199  < 2e-16 ***
RACE2         458.2427  1085.2320   0.422    0.673
RACE3         330.5184  2629.5121   0.126    0.900
RACE4        -499.3818  1520.9293  -0.328    0.743
RACE5       -1784.5776  1532.0048  -1.165    0.245
RACE6        -594.2921  1859.1271  -0.320    0.749
APRDRG         -7.8175     0.6881 -11.361  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2622 on 489 degrees of freedom
Multiple R-squared:  0.5544,    Adjusted R-squared:  0.5462
F-statistic:  67.6 on 9 and 489 DF,  p-value: < 2.2e-16

>
```

- Based on the output we can see that the Age and Length of stay affects the total Hospital cost.
- Cost is directly proportional to the Length i.e. higher the Length of stay of patients will result to higher hospital cost.
- As per the output we can see that with an increase of 1 day stay, the hospital cost will increase by 742.